Erdem Bıyık

https://ebiyik.github.io
https://scholar.google.com/citations?user=P-G3sjYAAAAJa

Email: biyik@usc.edu
1031 Downey Way, Ginsburg Hall (GCS)
Room 305C, Los Angeles, CA 90089

Current Position

- **University of Southern California** — Los Angeles, CA, USA
  Assistant Professor of Computer Science — *Aug 2023 – present*
  Assistant Professor of Electrical and Computer Engineering - Systems (by courtesy) — *Jan 2024 – present*

Education

- **Stanford University** — Stanford, CA, USA
  Doctor of Philosophy in Electrical Engineering; GPA: 4.07/4.00; Advisor: Prof. Dorsa Sadigh — *Sep 2017 – Jun 2022*
- **Stanford University** — Stanford, CA, USA
  Master of Science in Electrical Engineering; GPA: 4.08/4.00 — *Sep 2017 – Apr 2019*
- **Bilkent University** — Ankara, Türkiye
  Bachelor of Science in Electrical and Electronics Engineering; GPA: 4.00/4.00; Rank: 1/170 — *Aug 2012 – Jun 2017*
- **National University of Singapore** — Singapore
  Exchange Student in Electrical and Computer Engineering — *Aug 2015 – Dec 2015*

Work Experience

- **UC Berkeley, Center for Human-Compatible Artificial Intelligence (CHAI)** — Berkeley, CA, USA
  Postdoctoral Researcher — *Jul 2022 - Jul 2023*
  - **Reward Learning**: Working on various active reward learning projects for robotics with Prof. Stuart Russell and Prof. Anca Dragan
  - **Reinforcement Learning for Deep Brain Stimulation**: Working on the applications of reinforcement learning algorithms for deep brain stimulation under the supervision of Prof. Anca Dragan
- **Google Research** — Mountain View, CA, USA
  Research Intern — *Jun 2021 - Sep 2021*
  - **Recommender Systems**: Working on active preference elicitation for recommender systems under the supervision of Dr. Yinlam Chow, Dr. Mohammad Ghavamzadeh, and Prof. Craig Boutilier
  - **Preference-based Reinforcement Learning**: Working on relaxing initial state assumptions in preference-based reinforcement learning under the supervision of Dr. Yinlam Chow and Dr. Mohammad Ghavamzadeh
- **National Magnetic Resonance Research Center (UMRAM)** — Ankara, Türkiye
  Undergraduate Researcher & Research Intern — *Apr 2016 - Sep 2017*
  - **csMRI**: Developing compressed sensing methods for accelerated MRI under the supervision of Prof. Tolga Çukur
  - **SSFP Imaging**: Developing coil compression methods and artifact suppression techniques for balanced SSFP under the supervision of Prof. Tolga Çukur
- **ASELSAN** — Ankara, Türkiye
  Intern (2015), Research Engineer (2017) — *Jun 2015 - Jul 2015, Apr 2017 - Aug 2017*
  - **Military Communications (2015)**: Developing a C++ program to decode and encode data in MIL-STD-3014 protocol with an easy-to-use interface
  - **Algorithms Design (2015)**: Designing algorithms to solve composite launch acceptability region problem
  - **SSFP Imaging (2017)**: R&D projects related to field inhomogeneity, banding profile estimation, coil compression and banding suppression in bSSFP MRI under the supervision of Dr. Aykut Koç
- **École polytechnique fédérale de Lausanne (EPFL)** — Lausanne, Switzerland
  Research Intern via Summer@EPFL Program — *Jun 2016 - Sep 2016*
  - **Approximate Message Passing**: Research about approximate message passing algorithms as an intern in the Information Processing Group under the supervision of Prof. Rüdiger Urbanke
- **Anadolu Agency** — Ankara, Türkiye
  Intern — *Jun 2014 - Jul 2014*
  - **Image Processing**: Developing a Java, C++ and MySQL based program that uses image registration, matching and comparison techniques to detect copyright infringements

Teaching Experience

- **University of Southern California** — Los Angeles, CA
  Instructor — *Fall 2023, Spring 2024, Fall 2024, Fall 2025, Spring 2026*
  - **CSCI445L Introduction to Robotics (Spring 2024, Spring 2026)**: Teaching the course
  - **CSCI699 Robot Learning (Fall 2023, Fall 2024, Fall 2025)**: Designing and teaching the course
- **Stanford University** — Stanford, CA
  Teaching Assistant — *Winter 2020, Winter 2021*
  - **CS237B / EE260B / AA174B / AA274B Principles of Robot Autonomy II**: Holding office hours and sections, preparing and grading homeworks and exams
    Awarded "outstanding course assistant" by the CS department (top 5%) in Winter 2021
- **Bilkent University** — Ankara, Türkiye
  Teaching Assistant — *Spring 2014, Fall 2016*
  - **CS114 Introduction to Programming for Engineers (2014)**: Teaching in the weekly tutorials & recitations
  - **EEE211 Analog Electronics (2016)**: Teaching, interviewing and grading students in the laboratory sessions

Mentoring

- **Current Ph.D. Students**: Anthony Liang (co-advised with Stephen Tu), Pavel Czempin, Yigit Korkmaz, Yutai Zhou, Yimin Tang (co-advised with Sven Koenig), Ayano Hiranaka (co-advised with Daniel Seita), Bo-Ruei Huang
- **Current Master's Students**: Yusen Luo, Zhaoyang Li, Litian Gong, Abha Jha, Cagan Bakirci, Asmaa Hassan, Abhinav Srivatsa, Haobai Zhan, Ruopeng Huang, Aleyna Kara
- **Current Undergraduate Students**: Urvi Bhuwania, Ashley Lan, Kevin Kim, Arpita Tomar, Yasir White, Qingyang (Bob) Zeng, Alina Du, Jingzhen (Steve) Wang, Nitya Kashyap, Fatemeh Bahrani, Austin Tsai, Sharvil Oza, Lincoln Lam, Hugo Albert Bonet, Jamie Wei, Julia Chun
- **Ph.D. Alumni**
  - Jesse Zhang (2025, co-advised with Jesse Thomason and Joseph Lim). Next: Postdoc at UW.
    - Thesis Title: Scaling Robot Adaptation with Large Model Guidance
  - Ayush Jain (2024, co-advised with Joseph Lim). Next: Research Scientist at Meta.
    - Thesis Title: Decision Making in Complex Action Spaces
  - Sumedh Sontakke (2024, co-advised with Laurent Itti). Next: Applied Scientist at Amazon Robotics.
    - Thesis Title: Foundation Models for Embodied AI
- **Master's Alumni**
  - Matthew Hong (2025). Next: Research Fellow at the University of Washington.
    - USC Viterbi CS Best MS Research Award in 2025
  - Xinhu Li (2025). Next: Research Fellow at Tsinghua University.
  - Dhanush Varma (2025). Next: Ph.D. Student in Computer Science at University of Colorado Boulder.
  - Mihir Kulkarni (2025). Next: Ph.D. Student in Computer Science at UT Dallas.
  - Zhaojing Yang (2024). Next: Ph.D. Student in Computer Science at UC San Diego.
  - Jiahui Zhang (2024). Next: Ph.D. Student in Computer Science at UT Dallas.
  - Thomas Reeves (2024). Next: Ph.D. Student in Computer Science at USC.
- **Undergraduate Alumni**
  - Jaiv Doshi (2025). Research Scientist at Tangible Robotics.
  - Shreyas Ramanujam (2025, Intern from IIT Madras). Next: Master's Student at Stanford University.
  - Eisuke Hirota (2025, Intern from SUNY Binghamton). Next: Researcher at MIT Lincoln Laboratory.
  - Miru Jun (2024). Next: Data Engineer at US Navy.

Honors & Awards

- **2026 Office of Naval Research (ONR) Young Investigator Program (YIP) Award**: For project titled "Learning Dexterous Robot Hand Behaviors from Multimodal Human Feedback"
- **KAUST Rising Star in AI**: Invited to the KAUST Rising Stars in AI Symposium 2026 to give a talk
- **RLC 2025 Outstanding Paper Award on Empirical Reinforcement Learning Research**: For the paper "Mitigating Suboptimality of Deterministic Policy Gradients in Complex Q-functions"
- **Top Reviewer at UAI 2025**: Recognition given by the program committee
- **Best Paper Award at the "2nd Workshop on Out-of-Distribution Generalization in Robotics: Towards**

Reliable Learning-Based Autonomy" at RSS 2025: For the paper "ReWiND: Language-Guided Rewards Teach Robot Policies without New Demonstrations". Awarded to the top paper among ∼50 papers.
- **2024 Outstanding Paper Finalist at TMLR**: For the paper "Open Problems and Fundamental Limitations of Reinforcement Learning from Human Feedback". Awarded to 5 papers.
- **Stanford CS Outstanding Course Assistant**: For the "Principles of Robot Autonomy II" course in Winter 2021. Awarded to the top 5% of CAs in the department.
- **Qualcomm Innovation Fellowship 2020 North America**: Finalist (as a group of two PhD students)
- **HRI 2020 Honorable Mention in Technical Advances**: For the paper "When Humans Aren't Optimal: Robots that Collaborate with Risk-Aware Humans". Awarded to 3 papers in the Technical Advances track.
- **Qualcomm Innovation Fellowship 2019 North America**: Finalist (as a group of two PhD students)
- **Stanford University James D. Plummer Graduate Fellowship (2017-2018)**: Full tuition waiver & stipend during the first year of PhD program
- **Bilkent University Electrical and Electronics Engineering Department, Graduation Awards (2017)**: Academic Excellence, Research Excellence, Voluntary Professional Activities, Social Awareness and Activities
- **Scholarship of the Turkish Prime Ministry (2012-2017)**: Monthly stipend during the BSc program, for ranking in top 100 among 1.8 million students in nationwide university entrance exam
- **Bilkent University Comprehensive Scholarship (2012-2017)**: Full tuition waiver & stipend during the BSc program
- **IEEExtreme Programming Competitions**: 2nd in Türkiye, 78th among all participants, 2015. 1st in Türkiye, 73rd among all participants, 2014. 1st in Türkiye, 100th among all participants, 2013 (All as a group of three students)
- **Turkish Intelligence Foundation (TZV) Marathon**: Ranked twice in top 25 and once 6th, 2012-2014
- **İşbank Golden Youth Award (2012)**: For outstanding performance in the nationwide university entrance exam
- **Nationwide University Entrance Exam (LYS)**: Ranked 13th among 1.8 million students in Türkiye, 2012

Invited Talks & Poster Presentations

- Maximally Informative, Minimally Demanding: Learning from Human Feedback
  - KAUST Rising Stars in AI Symposium 2026
  - UCLA, Computer Science, Department Seminar (2025)
  - Carnegie Mellon University, Guest Lecture at "Human-Robot Interaction" (2025)
  - Inha University, Department of Aerospace Engineering (2025)
  - CoRL 2025 Workshop on Resource-Rational Robot Learning
- Scaling Robotic Reward Foundation Models with Automatically Curated Supervision
  - Apple Robotics (2026)
- SPUR: Scaling Reward Learning from Human Demonstrations
  - CoRL 2025 Workshop on Evaluation and Deployment Across the Robot Learning Lifecycle (Eval&Deploy)
- Robots That Pay Attention, Follow Instructions, and Learn Stably
  - Sony AI (2025)
- Robot Learning with Minimal Human Feedback
  - Advancing AI Capacity in Indonesian Universities: A Collaborative Workshop for Faculty Members (2025)
  - Stanford University, Guest Lecture at "Principles of Robot Autonomy II" (2025)
  - University of Pennsylvania, General Robotics, Automation, Sensing, & Perception Lab (GRASP) Seminar (2025)
  - UC Santa Barbara, The Center for Control, Dynamical Systems, and Computation Seminar (2025)
  - Bilkent University, Electrical and Electronics Engineering Department, Graduate Seminar (2024)
  - University of Washington, Robotics Seminar (2024)
  - University of Illinois Urbana-Champaign, Human-Centered Autonomy Lab (2024)
- Learning Humans' Tradeoffs between Objectives via Language Feedback
  - RSS 2025 Workshop on Multi-Objective Optimization and Planning in Robotics
- Multi-Agent Inverse Q-Learning from Demonstrations
  - ICRA 2025
- RAILGUN: A Unified Convolutional Policy for Multi-Agent Path Finding Across Different Environments and Tasks
  - ICRA 2025 Workshop on Foundation Models and Neuro-Symbolic AI for Robotics (poster)
- Asking Easy Questions: A User-Friendly Approach to Active Reward Learning

    - Inzva & AI Safety Türkiye, AI Safety Talks (2025)
    - CoRL 2019 (poster)
- Open Problems in Reinforcement Learning from Human Feedback and Potential Solutions for Data-Efficiency
    - Caltech AI Alignment Club (2025)
    - University of Amsterdam, The Amsterdam Machine Learning Lab (2023)
- Efficient Robot Learning via Interaction with Humans
    - New Faculty Highlights at the 39th Annual AAAI Conference on Artificial Intelligence, 2025
    - Southern California Robotics (SCR) Symposium 2023
- Reinforcement Learning from Comparative Language Feedback
    - The Inaugural Conference of the International Association for Safe and Ethical AI (IASEAI 2025 – poster)
- Trajectory Improvement and Reward Learning from Comparative Language Feedback
    - CoRL 2024 (poster)
- Preference Learning from Minimal Human Feedback for Interactive Autonomy
    - ODSC West 2024 Conference (Open Data Science Conference)
    - International Conference on Micro, Nano and Smart Systems (IC-MNSS) 2024
    - IIT Hyderabad, Mechanical and Aerospace Engineering, Department Seminar & Technology Innovation Hub on Autonomous Navigation and Data Acquisition Systems (2024)
- Making Robot Learning More Natural Through Human Saliency and Language
    - RSS 2024 Workshop on "Mechanisms for Mapping Human Input to Robots: From Robot Learning to Shared Control/Autonomy"
- Data- and Time-Efficient Learning of Human Preferences
    - 8th Annual Center for Human-Compatible Artificial Intelligence (CHAI) Workshop (2024)
- Optimizing Robot Behavior via Comparative Language Feedback
    - Human-Interactive Robot Learning (Workshop at HRI 2024 – poster)
- Data-Efficient Learning from Human Feedback and Large Pretrained Models for Robotics
    - Arizona State University, School of Computing and Augmented Intelligence (2023)
- ViSaRL: Visual Reinforcement Learning Guided by Human Saliency
    - 7th Annual Center for Human-Compatible Artificial Intelligence (CHAI) Workshop (2023 – poster)
- Learning Preferences for Interactive Autonomy
    - Middle East Technical University, Robotics and AI Technologies Application and Research Center (2022)
    - Sonoma State University, Engineering Colloquium (2022)
    - Cornell University, Robotics Seminar (2022)
    - University of Wisconsin-Madison, Computer Science, Department Seminar (2022)
    - University of Illinois Urbana-Champaign, Computer Science, Department Seminar (2022)
    - University of Michigan, Robotics Institute, Department Seminar (2022)
    - University of Southern California, Computer Science, Department Seminar (2022)
    - Imperial College London, Department of Aeronautics, Aerodynamics & Control Seminar (2022)
    - Carnegie Mellon University, Robotics Institute, Department Seminar (2022)
    - UCLA, Electrical and Computer Engineering, Department Seminar (2022)
    - Bilkent University, Electrical and Electronics Engineering, Graduate Seminar (2021)
    - UC Berkeley, Center for Human-Compatible Artificial Intelligence (CHAI), Beneficial AI Seminar (2021)
    - Sabancı University, Computer Science & Engineering, Department Seminar (2021)
    - Koç University, AI Meetings (2021)
    - Bay Area Robotics Symposium (BARS) 2021 (poster)
    - Virginia Tech, Mechanical Engineering, Department Seminar (2021)
    - Caltech Yue Lab (2021)
    - UT Austin, Personal Autonomous Robotics Lab (2021)
    - UC Berkeley, InterACT Lab (2021)
- Learning Multimodal Rewards from Rankings
    - 6th Annual Center for Human-Compatible Artificial Intelligence (CHAI) Workshop (2022)

- CoRL 2021
- Learning from Human Preferences
  - Stanford University, Guest Lecture at "Safe and Interactive Robotics" (2022)
- Behavior Modeling, via Reward Learning and for Autonomous Driving
  - Nuro (2022)
- Learning from Humans for Adaptive Interaction
  - HRI Pioneers 2022
- APReL: A Library for Active Preference-based Reward Learning Algorithms
  - Human-Interactive Robot Learning (HIRL) (Workshop at HRI 2022)
  - HRI 2022
  - AI-HRI 2021 at AAAI Fall Symposium Series
- Partner-Aware Algorithms in Decentralized Cooperative Bandit Teams
  - AAAI 2022
  - AI-HRI 2021 at AAAI Fall Symposium Series
- The Role of Representations in Human-Aware Learning and Control *(with Dorsa Sadigh)*
  - "Aware-Learning: How to Benefit from Priors" Workshop @ CDC 2021
- Interactive Robotics through the Lens of Learning *(with Dorsa Sadigh)*
  - NCCR (The National Centre of Competence in Research, Switzerland) Automation Seminar, 2021
- Leveraging Smooth Attention Prior for Multi-Agent Trajectory Prediction
  - Center for Automotive Research at Stanford (CARS) Annual Meeting (2021 – poster)
- Learning Reward Functions from Scale Feedback
  - CoRL 2021 (poster)
- Learning how to Dynamically Route Autonomous Vehicles on Shared Roads
  - ETH Zurich, Institute for Dynamic Systems and Control, Autonomy Talks (2021)
  - The 32nd IEEE Intelligent Vehicles Symposium Workshop (2021)
  - 3rd NorCal Control Workshop (2021)
  - Stanford Robotics Lunch (2019)
  - Center for Automotive Research at Stanford (CARS) Annual Meeting (2020 – poster)
- Walking the Boundary of Learning and Interaction *(with Dorsa Sadigh)*
  - 3rd Robot Learning Workshop: Grounding Machine Learning Development in the Real World @ NeurIPS 2020
- When Humans Aren't Optimal: Robots that Collaborate with Risk-Aware Humans *(with Minae Kwon)*
  - University of Chicago, Graduate Seminar on Topics in Human-Robot Interaction (2020)
- Active Preference-Based Gaussian Process Regression for Reward Learning
  - RSS 2020
- The Green Choice: Learning and Influencing Human Decisions on Shared Roads
  - CDC 2019
- Active Learning of Reward Dynamics from Hierarchical Queries
  - IROS 2019
- Efficient and Safe Exploration in Deterministic Markov Decision Processes with Unknown Transition Models
  - ACC 2019
  - Stanford AI Safety Retreat 2019 (poster)
- Batch Active Preference-Based Learning of Reward Functions
  - CoRL 2018
  - Stanford HAI 2019 (poster)
  - TRI Joint University Workshop 2019 (poster)
  - Stanford SystemX Fall 2018 (poster)
  - BARS - Bay Area Robotics Symposium (BARS) 2018 (poster)

Professional Activities

- **Conferences Organized**
  - **Local Arrangements Co-Chair**, Robotics: Science and Systems (RSS), 2025
  - **Organizer**, Human-in-the-Loop Robot Learning: Teaching, Correcting, and Adapting (Workshop at Robotics: Science and Systems (RSS)), 2025
  - **Finance Chair**, Conference on Robot Learning (CoRL), 2024
  - **Local Arrangements Chair**, Bay Area Robotics Symposium (BARS), 2021
  - **Local Arrangements Chair**, Bay Area Robotics Symposium (BARS), 2020
  - **Organizer**, Emergent Behaviors in Human-Robot Systems (Workshop at Robotics: Science and Systems (RSS)), 2020
  - **Local Arrangements Chair**, Bay Area Robotics Symposium (BARS), 2018
- **Program Committee Member**
  - AAAI 2026 New Faculty Highlights Track
  - ACM/IEEE International Conference on Human-Robot Interaction (HRI), 2025
  - The 39th Annual AAAI Conference on Artificial Intelligence, AI Alignment Track (2025)
  - 8th Annual Center for Human-Compatible Artificial Intelligence (CHAI) Workshop, 2024
  - 7th Annual Center for Human-Compatible Artificial Intelligence (CHAI) Workshop, 2023
- **Editorial Board**
  - **Associate Editor** for Robot Learning, IEEE Robotics and Automation Letters (RA-L), 2023 – present
  - **Area Chair** for Robotics: Science and Systems 2026
  - **Area Chair** for Late Breaking Results at the ACM/IEEE International Conference on Human-Robot Interaction (HRI), 2025–2026
  - **Associate Editor**, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2024–2025
  - **Area Chair** for Robotics: Science and Systems - Pioneers Workshop 2025
  - **Area Chair**, Responsible AI (RAI) (Workshop at International Conference on Learning Representations (ICLR)), 2021
- **Panelist**
  - RSS 2025 Workshop on Multi-Objective Optimization and Planning in Robotics
  - Human Value Learning Panel at the 8th Annual Center for Human-Compatible Artificial Intelligence (CHAI) Workshop, 2024
- **Reviewer for Journal Papers**
  - ACM Transactions on Autonomous and Adaptive Systems (TAAS)
  - ACM Transactions on Human-Robot Interaction (THRI)
  - ACM Transactions on Multimedia Computing, Communications, and Applications (TOMM)
  - Artificial Intelligence
  - Autonomous Agents and Multi-Agent Systems
  - Autonomous Robots (AURO)
  - Cognitive Systems Research
  - Frontiers in Computer Vision
  - Frontiers in Robotics and AI
  - IEEE Control System Letters (L-CSS)
  - IEEE Robotics and Automation Letters (RA-L)
  - IEEE Robotics and Automation Magazine (RAM)
  - IEEE Transactions on Artificial Intelligence (TAI)
  - IEEE Transactions on Automatic Control (TAC)
  - IEEE Transactions on Automation Science and Engineering (T-ASE)
  - IEEE Transactions on Cognitive and Developmental Systems (TCDS)
  - IEEE Transactions on Control Systems Technology (TCST)
  - IEEE Transactions on Human-Machine Systems (THMS)
  - IEEE Transactions on Intelligent Transportation Systems (T-ITS)
  - IEEE Transactions on Knowledge and Data Engineering (TKDE)

- IEEE Transactions on Networking (TNET)
- IEEE Transactions on Neural Networks and Learning Systems (TNNLS)
- IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
- IEEE Transactions on Robotics (T-RO)
- IEEE Transactions on Systems, Man and Cybernetics: Systems (TSMC-S)
- IEEE/ASME Transactions on Mechatronics (TMECH)
- International Journal of Machine Learning and Cybernetics
- International Journal of Robotics Research (IJRR)
- Journal of Machine Learning Research (JMLR)
- MDPI Robotics
- Nature Communications
- Proceedings of the Royal Society A
- Robotica

- **Reviewer for Conference Papers**
  - International Conference on Learning Representations (ICLR), 2025-2026
  - IEEE International Conference on Robotics and Automation (ICRA), 2021–2024, 2026
  - Conference on Robot Learning (CoRL), 2019–2025
  - IEEE Conference on Decision and Control (CDC), 2019–2020, 2023–2025
  - Robotics: Science and Systems (RSS), 2020–2021, 2023–2025
  - Conference on Uncertainty in Artificial Intelligence (UAI), 2023–2025 *(Top Reviewer Recognition in 2025)*
  - International Conference on Machine Learning (ICML), 2025
  - International Conference on Artificial Intelligence and Statistics (AISTATS), 2025
  - Conference on Neural Information Processing Systems (NeurIPS), 2023–2024
  - IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2020, 2022–2024
  - IEEE International Conference on Robot & Human Interactive Communication (RO-MAN), 2024
  - Learning for Dynamics & Control (L4DC), 2020, 2023–2024
  - ACM/IEEE International Conference on Human-Robot Interaction (HRI), 2020–2024
  - International Joint Conference on Artificial Intelligence (IJCAI), 2023
  - American Control Conference (ACC), 2019, 2021–2022
  - IEEE Conference on Control Technology and Applications (CCTA), 2021
  - IEEE-RAS International Conference on Humanoid Robots (Humanoids), 2020
  - International Conference on Computer-Aided Verification (CAV), 2019
  - ACM International Conference on Hybrid Systems: Computation and Control (HSCC), 2019
- **Reviewer for Workshop Proposals**
  - International Conference on Learning Representations (ICLR), 2026
- **Reviewer for Workshop Papers**
  - RSS Pioneers Workshop 2023–2026
  - 2nd Workshop on Semantic Reasoning and Goal Understanding in Robotics (SemRob) at RSS 2025
  - Workshop on User-Aligned Assessment of Adaptive AI Systems (AIA) at IJCAI 2025
  - HRI Pioneers Workshop 2023–2024
  - Interactive Learning with Implicit Human Feedback Workshop (ILHF) at ICML 2023
  - AAMAS Workshop on Rebellious and Disobedient AI (RaD-AI), 2022–2023
  - CoRL Workshop on Aligning Robot Representations with Humans, 2022
  - NeurIPS Workshop on Foundation Models for Decision Making (FMDM), 2022
  - NeurIPS Workshop on Progress and Challenges in Building Trustworthy Embodied AI (TEA), 2022
  - NeurIPS Workshop on Machine Learning Safety, 2022
  - RSS Workshop on Learning from Diverse, Offline Data (L-DOD), 2022
  - RSS Workshop on Social Intelligence in Humans and Robots, 2022
  - Artificial Intelligence for Human-Robot Interaction Symposium (AI-HRI) at AAAI Fall Symposium Series, 2021
  - ICRA Workshop on Social Intelligence in Humans and Robots, 2021
  - Bridging AI and Cognitive Science (BAICS) (Workshop at International Conference on Learning Repre-

sentations (ICLR)), 2020

- **Session Chair at Conferences**
  - IROS 2025
  - RSS 2024
  - ICMNSS 2024
  - ICRA 2024

Committee Membership for Ph.D. Students

- **University of Southern California, Computer Science**
  - Jeremy Morgan, Qualifying Exam (Nov. 2023), Thesis Proposal (Apr. 2026)
  - Anthony Liang, Qualifying Exam (Oct. 2024), Thesis Proposal (Oct. 2025), Defense (Apr. 2026)
  - Sophie Hsu, Qualifying Exam (Dec. 2024), Thesis Proposal (Dec. 2025), Defense (Apr. 2026)
  - Saeed Hedayatian, Qualifying Exam (Apr. 2026)
  - Yiyang Ling, Qualifying Exam (Apr. 2026)
  - Satyajeet Das, Qualifying Exam (Apr. 2026)
  - Yunshuang Li, Qualifying Exam (Apr. 2026)
  - Shuqin Zhu, Thesis Proposal (Mar. 2026)
  - Yigit Korkmaz, Qualifying Exam (Mar. 2026)
  - I-Chun Liu, Qualifying Exam (Mar. 2025), Thesis Proposal (Feb. 2026)
  - Shipeng Liu, Defense (Feb. 2026)
  - Yimin Tang, Qualifying Exam (Nov. 2025)
  - Hamed Alimohammadzadeh, Qualifying Exam (Nov. 2025)
  - Robby Costales, Qualifying Exam (Sep. 2024), Thesis Proposal (Apr. 2025), Defense (Nov. 2025)
  - Grace Zhang, Qualifying Exam (May 2024), Thesis Proposal (Apr. 2025), Defense (Nov. 2025)
  - Sumeet Batra, Defense (Nov. 2025)
  - Varun Sreedhara Bhatt, Qualifying Exam (Feb. 2024), Thesis Proposal (Oct. 2025)
  - Zhonghao Shi, Qualifying Exam (Jun. 2024), Thesis Proposal (Jun. 2025)
  - Jesse Zhang, Thesis Proposal (Sep. 2024), Defense (May 2025)
  - Bingjie Tang, Thesis Proposal (May 2024), Defense (May 2025)
  - Kegan Strawn, Qualifying Exam (Oct. 2024), Thesis Proposal (May 2025)
  - Bryon Tjanaka, Thesis Proposal (Apr. 2025)
  - Shihan Zhao, Qualifying Exam (Apr. 2025)
  - Soumyaroop Nandi, Thesis Proposal (Apr. 2025)
  - Anand Balakrishnan, Thesis Proposal (Dec. 2024)
  - Ayush Jain, Thesis Proposal (Feb. 2024), Defense (Nov. 2024)
  - Kiran Lekkala, Defense (Nov. 2024)
  - Xin Zhu, Thesis Proposal (Jun. 2024)
  - Sheryl Paul, Qualifying Exam (May 2024)
  - Sumedh Sontakke, Thesis Proposal (Feb. 2024), Defense (Apr. 2024)
  - Yang Chen, Qualifying Exam (Mar. 2024)
  - Weizhe Chen, Qualifying Exam (Mar. 2024)
- **University of Southern California, Electrical and Computer Engineering**
  - Yue Hu, Defense (Apr. 2026)
  - Armin Abdollahi, Qualifying Exam (Jan. 2026)
  - Yusuf Umut Ciftci, Qualifying Exam (Jan. 2026)
  - Eray Erturk, Qualifying Exam (Nov. 2025)
  - Mingye Li, Qualifying Exam (Jan. 2025), Defense (Aug. 2025)
  - Javier Borquez, Defense (Jun. 2025)
  - Yuning Zhang, Qualifying Exam (May 2025)
  - Zeming Cheng, Qualifying Exam (Sep. 2024), Defense (Apr. 2025)
  - Mustafa Altay Karamuftuoglu, Qualifying Exam (May 2024), Defense (Apr. 2025)

Outreach and Mentoring

- **USC Thomas Lord Department of Computer Science, 2024-present** Los Angeles, CA, USA
  - **Undergraduate Mentoring Program Organizer**: Organizing the program where PhD students mentor undergraduates about their future career, getting involved in CS research, and applying for industry or graduate school
- **USC Robotics and Autonomous Systems Center (RASC), 2024-present** Los Angeles, CA, USA
  - **Faculty Mentor for USC Robotics Seminar (URoS)**: Mentoring Ph.D. students in running a biweekly robotics seminar to host students presentations and invited speakers
  - **Faculty Mentor for Robotics and Autonomous Systems Center (RASC) Blog**: Mentoring Ph.D. students in running a blog website about the robotics and autonomous systems resarch at USC
- **USC Viterbi School of Engineering, K-12 STEM Center, 2025** Los Angeles, CA, USA
  - **Instructor at CS@SC Coding Camps**: Teaching middle school and high school students "how robots learn to imitate humans" in two sessions
- **Berkeley Artificial Intelligence Safety Initiative for Students, 2023-2024** Berkeley, CA, USA
  - **Mentor at Supervised Program for Alignment Research**: Mentoring three undergraduate students on a multi-agent inverse reinforcement learning research project
- **Berkeley Artificial Intelligence Research Lab (BAIR), 2022-2023** Berkeley, CA, USA
  - **BAIR Undergraduate Mentoring Program Mentor**: Mentoring undergraduate students about their future career, getting involved in AI research, and applying for industry or graduate school
- **Stanford University Department of Computer Science, 2020-2022** Stanford, CA, USA
  - **CS Mentorship Program Organizer**: Organizing the CS mentorship program and the related social events
  - **CS Mentorship Program Mentor**: Mentoring first and second-year undergraduate students about their future career, getting involved in research projects, and applying for industry or graduate school
- **Stanford University AI Laboratory (SAIL), 2018-2020** Stanford, CA, USA
  - **AI Mentorship Program Organizer**: Organizing the AI mentorship program and the related social events
  - **AI Mentorship Program Mentor**: Mentoring first and second-year undergraduate students about their future career, getting involved in AI research, and applying for industry or graduate school
  - **Robotics Lunch Organizer (2019-2020)**: Organizing bi-weekly robotics lunch sessions where invited professors, postdoctoral researchers and Ph.D. candidates present their research
- **Stanford STEM to SHTEM Summer Internship Program, 2019** Stanford, CA, USA
  - **Mentor**: Mentoring high school students in a research project on decision making under risk / time constraints
- **IEEE Bilkent Student Branch, 2012-2017** Ankara, Türkiye
  - **Road to University Volunteer (2013-2017)**: Introducing engineering and campus life to high school students from all around Türkiye
  - **Vice Chair (2014-2015)**: Managing and organizing the technical trainings and social events
  - **Robotics and Automation Society Member (2012-2015)**: Assistantship in the electronics and robotics focused tutorials
  - **Web Team Member (2012-2014) and Webmaster (2013-2014)**: Designing and managing the website of IEEE Bilkent SB, its communities and events; teaching in web design tutorials and in MATLAB tutorials; organizing a Java programming competition and weekly brain teasers
- **GazeteBilkent, 2014-2017** Ankara, Türkiye
  - **Online Operations Manager**: Managing the website of GazeteBilkent, an online newspaper
- **Bilkent University, 2014-2017** Ankara, Türkiye
  - **Webmaster of Electrical and Electronics Engineering (2013-2017), Mechanical Engineering (2015-2017), Economics (2016-2017) Departments**: Designing, developing and managing the websites of the departments and developing necessary web tools
  - **1st and 2nd Industrial Design Projects Fairs Student Coordinator and Webmaster (2015, 2016)**: Organizing the fair in which the senior students of Bilkent EEE present their industry projects
  - **Graduate Research Conference '15 Student Coordinator (2015)**: Organizing the conference series in which graduate students of Bilkent EEE present their research and projects
  - **Graduate Research Conference '14 Webmaster and Organization Team Member (2014)**: Organizing the conference series in which graduate students of Bilkent EEE present their research and projects
- **Bilkent Chess Society, 2013-2014** Ankara, Türkiye
  - **Vice Chair**: Organizing tournaments on campus, participating in local events as a team
- **Bilkent Academic Career Club, 2013** Ankara, Türkiye
  - **Physics Olympiads Volunteer**: Organizing a physics competition among high school students

Patents

1. Z Cao, **E Bıyık**, WZ Wang, A Raventos, A Gaidon, G Rosman, D Sadigh. "Reinforcement Learning Based Control of Imitative Policies for Autonomous Driving", US Patent Application No. US17/002,650.

Journal Publications

1. K Lekkala, H Bao, SA Sontakke, **E Bıyık**, L Itti. "Value Explicit Pretraining for Learning Transferable Representations", IEEE Robotics and Automation Letters (RA-L), 2026.
2. K Baraka, I Idrees, TK Faulkner, **E Bıyık**, S Booth, M Chetouani, DH Grollman, A Saran, E Senft, S Tulli, A Vollmer, A Andriella, H Beierling, T Horter, J Kober, I Sheidlower, ME Taylor, SV Waveren, X Xiao. "Human-Interactive Robot Learning: Definition, Challenges, and Recommendations", ACM Transactions on Human Robot Interaction (THRI), 2025.
3. E Hirota*, Z Yang*, A Hiranaka, M Jun, J Tien, SJ Russell, A Dragan, **E Bıyık** (*equal contribution). "Active Reward Learning and Iterative Trajectory Improvement from Comparative Language Feedback", The International Journal of Robotics Research (IJRR), 2025; doi:10.1177/02783649251394742.
4. **E Bıyık**. "Training Robots with Natural and Lightweight Human Feedback", AI Magazine, 2025.
5. **E Bıyık**, N Anari, D Sadigh. "Batch Active Learning of Reward Functions from Human Preferences", ACM Transactions on Human-Robot Interaction (THRI), 2024; doi:10.1145/3649885.
6. S Casper*, X Davies*, C Shi, TK Gilbert, J Scheurer, J Rando, R Freedman, T Korbak, D Lindner, P Freire, T Wang, S Marks, CR Segerie, M Carroll, A Peng, P Christoffersen, M Damani, S Slocum, U Anwar, A Siththaranjan, M Nadeau, EJ Michaud, J Pfau, D Krasheninnikov, X Chen, L Langosco, P Hase, **E Bıyık**, A Dragan, D Krueger, D Sadigh, D Hadfield-Menell (*equal contribution). "Open Problems and Fundamental Limitations of Reinforcement Learning from Human Feedback", Transactions on Machine Learning Research (TMLR), 2023. ***Outstanding paper finalist.***
7. **E Bıyık**, N Huynh, MJ Kochenderfer, D Sadigh. "Active Preference-Based Gaussian Process Regression for Reward Learning and Optimization", The International Journal of Robotics Research (IJRR), 2023; doi: 10.1177/02783649231208729.
8. M Tucker, K Li, E Novoseller, M Pétriaux, G Burger, **E Bıyık**, M Masselin, D Sadigh, JW Burdick, Y Yue, AD Ames. *Anonymous Submission*, Nature Machine Intelligence, 2022. *(Submitted)*
9. **E Bıyık**, DP Losey, M Palan, NC Landolfi, G Shevchuk, D Sadigh. "Learning Reward Functions from Diverse Sources of Human Feedback: Optimally Integrating Demonstrations and Preferences", The International Journal of Robotics Research (IJRR), 2022; doi:10.1177/02783649211041652
10. DA Lazar*, **E Bıyık***, D Sadigh, R Pedarsani (*equal contribution). "Learning How to Dynamically Route Autonomous Vehicles on Shared Roads", Transportation Research Part C: Emerging Technologies (TR_C), 2021; doi: 10.1016/j.trc.2021.103258
11. **E Bıyık***, DA Lazar*, R Pedarsani, D Sadigh (*equal contribution). "Incentivizing Efficient Equilibria in Traffic Networks with Mixed Autonomy", IEEE Transactions on Control of Network Systems (TCNS), 2021; doi: 10.1109/TCNS.2021.3084045.
12. **E Biyik***, K Keskin*, SUH Dar, A Koc, T Çukur (*equal contribution). "Factorized sensitivity estimation for artifact suppression in phase-cycled bSSFP MRI", NMR in Biomedicine, 2020; doi: 10.1002/nbm.4228.
13. **E Biyik**, E Ilicak, T Çukur. "Reconstruction by Calibration over Tensors for Multi-Coil Multi-Acquisition Balanced SSFP Imaging", Magnetic Resonance in Medicine (MRM), 2017; doi: 10.1002/mrm.26902.
14. E Ilicak, LK Senel, **E Biyik**, T Çukur. "Profile-encoding reconstruction for multiple-acquisition balanced steady-state free precession imaging", Magnetic Resonance in Medicine (MRM), 2016; doi: 10.1002/mrm.26507.

Refereed Conference Publications

15. A Liang*, Y Korkmaz*, J Zhang, M Hwang, A Anwar, S Kaushik, A Shah, AS Huang, L Zettlemoyer, D Fox, Y Xiang, A Li, A Bobu, A Gupta, S Tu†, **E Bıyık**†, J Zhang† (*equal contribution, †equal advising). *Anonymous Submission*, Robotics: Science and Systems (RSS), Sydney, New South Wales, Australia, Jul. 2026. *(Submitted)*
16. R Huang, B Yang, W Gui, J Morgan, **E Bıyık**, J Li. *Anonymous Submission*, Robotics: Science and Systems (RSS), Sydney, New South Wales, Australia, Jul. 2026. *(Submitted)*
17. Y Zhou, **E Bıyık**. *Anonymous Submission*, 43rd International Conference on Machine Learning (ICML), Seoul, South Korea, Jul. 2026. *(Submitted)*
18. A Hiranaka, YC Hsu, **E Bıyık**†, D Seita† (†equal advising). *Anonymous Submission*, 43rd International Conference on Machine Learning (ICML), Seoul, South Korea, Jul. 2026. *(Submitted)*
19. P Czempin, T Trinh, MH Danesh, KX Nguyen, **E Bıyık**, B Plaut. *Anonymous Submission*, 43rd International Conference on Machine Learning (ICML), Seoul, South Korea, Jul. 2026. *(Submitted)*
20. Y Tang, S Koenig, **E Bıyık**. *Anonymous Submission*, 17th World Symposium on the Algorithmic Foundations of Robotics (WAFR), Oulu, Finland, Jun. 2026. *(Submitted)*
21. R Zhang, X Zhu, MP Khotbehsara, W Dao, **E Bıyık**, H Culbertson. *Anonymous Submission*, 34th ACM International

Conference on User Modeling, Adaptation and Personalization (UMAP), Gothenburg, Sweden, Jun. 2026. *(Submitted)*

22. Z Li, Z Ling, Y Zhou, L Gong, **E Bıyık**, H Su. "ORIC: Benchmarking Object Recognition under Contextual Incongruity in Large Vision-Language Models", 2026 IEEE/CVF Conference on Computer Vision & Pattern Recognition (CVPR), Denver, Colorado, USA, Jun. 2026.
23. L Gong, F Bahrani, Y Zhou, A Banayeeanzade, J Li, **E Bıyık**. "AutoFocus-IL: VLM-based Saliency Maps for Data-Efficient Visual Imitation Learning without Extra Human Annotations", 2026 IEEE International Conference on Robotics and Automation (ICRA), Vienna, Austria, Jun. 2026.
24. M Hong*, A Liang*, KJ Kim, HB Rajaprakash, J Thomason†, **E Bıyık**†, J Zhang† (*equal contribution, †equal advising). "HAND Me the Data: Fast Robot Adaptation via Hand Path Retrieval", 2026 IEEE International Conference on Robotics and Automation (ICRA), Vienna, Austria, Jun. 2026.
25. Y Ling*, K Owalekar*, O Adesanya, **E Bıyık**, D Seita (*equal contribution). "IMPACT: Intelligent Motion Planning with Acceptable Contact Trajectories via Vision-Language Models", 2026 IEEE Interational Conference on Robotics and Automation (ICRA), Vienna, Austria, Jun. 2026.
26. J Zhang*, M Memmel*, K Kim, D Fox, F Ramos, J Thomason, **E Bıyık**, A Gupta†, A Li† (*equal contribution, †equal advising). "PEEK: Guiding and Minimal Image Representations for Zero-Shot Generalization of Robot Manipulation Policies", 2026 IEEE International Conference on Robotics and Automation (ICRA), Vienna, Austria, Jun. 2026.
27. X Li, A Jain, Z Yang, Y Korkmaz, **E Bıyık**. "When a Robot is More Capable than a Human: Learning from Constrained Demonstrators", 14th International Conference on Learning Representations (ICLR), Rio de Janeiro, Brazil, Apr. 2026.
28. M Hwang, Y Korkmaz, D Seita†, **E Bıyık**† (†equal advising). "Causally Robust Preference Learning with Reasons", 14th International Conference on Learning Representations (ICLR), Rio de Janerio, Brazil, Apr. 2026.
29. Y Korkmaz*, U Bhuwania*, A Jain†, **E Bıyık**† (*equal contribution, †equal advising). "Actor-Free Continuous Control via Structurally Maximizable Q-Functions", 39th Conference on Neural Information Processing Systems (NeurIPS), San Diego, California, USA, Dec. 2025.
30. A Banayeeanzade*, F Bahrani*, Y Zhou, **E Bıyık** (*equal contribution). "GABRIL: Gaze-Based Regularization for Mitigating Causal Confusion in Imitation Learning", IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Hangzhou, China, Oct. 2025.
31. Y Tang*, X Xiong*, J Xi, J Li, **E Bıyık**, S Koenig (*equal contribution). "RAILGUN: A Unified Convolutional Policy for Multi-Agent Path Finding Across Different Environments and Tasks", IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Hangzhou, China, Oct. 2025.
32. J Zhang*, Y Luo*, A Anwar*, SA Sontakke, JJ Lim, J Thomason, **E Bıyık**, J Zhang (*equal contribution). ReWiND: Language-Guided Rewards Teach Robot Policies without New Demonstrations, 9th Conference on Robot Learning (CoRL), Seoul, South Korea, Sep. 2025.
33. A Jain, N Kosaka, X Li, K Kim, **E Bıyık**, J Lim. "Mitigating Suboptimality of Deterministic Policy Gradients in Complex Q-functions", 2nd Reinforcement Learning Conference (RLC), Edmonton, Alberta, Canada, Aug. 2025. ***Outstanding Paper Award on Empirical Reinforcement Learning Research.***
34. A Cheng*, Y Ji*, Z Yang*, X Zou, J Kautz, **E Bıyık**, H Yin, S Liu, X Wang (*equal contribution). "NaVILA: Legged Robot Vision-Language-Action Model for Navigation", Robotics: Science and Systems (RSS), Los Angeles, California, USA, Jun. 2025; doi: 10.15607/rss.2025.xxi.018.
35. Y Korkmaz, **E Bıyık**. "MILE: Model-based Intervention Learning", 2025 IEEE International Conference on Robotics and Automation (ICRA), Atlanta, Georgia, USA, May 2025; doi: 10.1109/icra55743.2025.11128018.
36. N Haynam, A Khoja, D Kumar, V Myers, **E Bıyık**. "Multi-Agent Inverse Q-Learning from Demonstrations", 2025 IEEE International Conference on Robotics and Automation (ICRA), Atlanta, Georgia, USA, May 2025; doi: 10.1109/ICRA55743.2025.11128370.
37. A Liang, G Tennenholtz, C Hsu, Y Chow, **E Bıyık**, C Boutilier. "DynaMITE-RL: A Dynamic Model for Improved Temporal Meta-Reinforcement Learning", 38th Conference on Neural Information Processing Systems (NeurIPS), Vancouver, British Columbia, Canada, Dec. 2024.
38. H Jaggi*, KC Murali*, E Fleisig, **E Bıyık** (*equal contribution). "Accurate and Data-Efficient Toxicity Prediction when Annotators Disagree", 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP), Miami, Florida, USA, Nov. 2024; doi: 10.18653/v1/2024.emnlp-main.1221.
39. Z Yang, M Jun, J Tien, SJ Russell, A Dragan, **E Bıyık**. "Trajectory Improvement and Reward Learning from Comparative Language Feedback", 8th Conference on Robot Learning (CoRL), Munich, Germany, Nov. 2024.
40. J Zhang, M Heo, Z Liu, **E Bıyık**, J Lim, Y Liu, R Fakoor. "EXTRACT: Efficient Policy Learning by Extracting Transferable Robot Skills from Offline Data", 8th Conference on Robot Learning (CoRL), Munich, Germany, Nov. 2024.
41. A Liang, J Thomason, **E Bıyık**. "ViSaRL: Visual Reinforcement Learning Guided by Human Saliency", IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Abu Dhabi, UAE, Oct. 2024; doi: 10.1109/iros58592.2024.10801388.
42. Y Wang*, Z Sun*, J Zhang, Z Xian, **E Bıyık**, D Held†, Z Erickson† (*equal contribution, †equal advising). "RL-VLM-F: Reinforcement Learning from Vision Language Foundation Model Feedback", 41st International Conference on Machine Learning (ICML), Vienna, Austria, Jul. 2024.

43. M Pan*, ML Schrum*, V Myers, **E Bıyık**, A Dragan (*equal contribution). "Coprocessor Actor Critic: A Model-Based Reinforcement Learning Approach For Adaptive Brain Stimulation", 41st International Conference on Machine Learning (ICML), Vienna, Austria, Jul. 2024.
44. E Ellis, GR Ghosal, SJ Russell, A Dragan, **E Bıyık**. "A Generalized Acquisition Function for Preference-based Reward Learning", 2024 IEEE International Conference on Robotics and Automation (ICRA), Yokohama, Japan, May 2024; doi: 10.1109/icra57147.2024.10611472.
45. S Sontakke, J Zhang, S Arnold, K Pertsch, **E Bıyık**, D Sadigh, C Finn, L Itti. "RoboCLIP: One Demonstration is Enough to Learn Robot Policies", 37th Conference on Neural Information Processing Systems (NeurIPS), New Orleans, Louisiana, USA, Nov. 2023.
46. V Myers, **E Bıyık**, D Sadigh. "Active Reward Learning from Online Preferences", 2023 IEEE International Conference on Robotics and Automation (ICRA), London, United Kingdom, May 2023; doi: 10.1109/icra48891.2023.10160439.
47. M Srivastava, **E Bıyık**, S Mirchandani, ND Goodman, D Sadigh. "Assistive Teaching of Motor Control Tasks to Humans", 36th Conference on Neural Information Processing Systems (NeurIPS), New Orleans, Louisiana, USA, Nov. 2022.
48. E Brockbank, H Wang, J Yang, S Mirchandani, **E Bıyık**, D Sadigh, J Fan. "How do People Incorporate Advice from Artificial Agents when Making Physical Judgments?", 44th Annual Meeting of the Cognitive Science Society (CogSci), Toronto, Ontario, Canada, Jul. 2022.
49. Z Cao, **E Bıyık**, G Rosman, D Sadigh. "Leveraging Smooth Attention Prior for Multi-Agent Trajectory Prediction", 2022 IEEE International Conference on Robotics and Automation (ICRA), Philadelphia, Pennsylvania, USA, May 2022; doi: 10.1109/ICRA46639.2022.9811718.
50. **E Bıyık**, A Talati, D Sadigh. "APReL: A Library for Active Preference-based Reward Learning Algorithms", ACM/IEEE International Conference on Human-Robot Interaction (HRI), Sapporo, Hokkaido, Japan, Mar. 2022; doi: 10.1109/hri53351.2022.9889650.
51. **E Bıyık**, A Lalitha, R Saha, A Goldsmith, D Sadigh. "Partner-Aware Algorithms in Decentralized Cooperative Bandit Teams", Thirty-Sixth AAAI Conference on Artificial Intelligence (AAAI), Vancouver, British Columbia, Canada, Feb. 2022; doi: 10.1609/aaai.v36i9.21158.
52. N Wilde*, **E Bıyık***, D Sadigh, SL Smith (*equal contribution). "Learning Reward Functions from Scale Feedback", 5th Conference on Robot Learning (CoRL), London, United Kingdom, Nov. 2021.
53. V Myers, **E Bıyık**, N Anari, D Sadigh. "Learning Multimodal Rewards from Rankings", 5th Conference on Robot Learning (CoRL), London, United Kingdom, Nov. 2021.
54. WZ Wang*, M Beliaev*, **E Bıyık***, DA Lazar, R Pedarsani, D Sadigh (*equal contribution). "Emergent Prosociality in Multi-Agent Games Through Gifting", 30th International Joint Conference on Artificial Intelligence (IJCAI), Montreal, Quebec, Canada, Aug. 2021; doi: 10.24963/ijcai.2021/61.
55. K Li, M Tucker, **E Bıyık**, E Novoseller, JW Burdick, Y Sui, D Sadigh, Y Yue, AD Ames. "ROIAL: Region of Interest Active Learning for Characterizing Exoskeleton Gait Preference Landscapes", 2021 IEEE International Conference on Robotics and Automation (ICRA), Xi'an, China, May 2021; doi: 10.1109/ICRA48506.2021.9560840.
56. M Beliaev, **E Bıyık**, DA Lazar, WZ Wang, D Sadigh, R Pedarsani. "Incentivizing Routing Choices for Safe and Efficient Transportation in the Face of the COVID-19 Pandemic", 12th ACM/IEEE International Conference on Cyber-Physical Systems (ICCPS), Nashville, Tennessee, USA, May 2021; doi: 10.1145/3450267.3450546.
57. Z Zhu, **E Bıyık**, D Sadigh. "Multi-Agent Safe Planning with Gaussian Processes", IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Las Vegas, Nevada, USA, Oct. 2020; doi: 10.1109/IROS45743.2020.9341169.
58. **E Bıyık***, N Huynh*, MJ Kochenderfer, D Sadigh (*equal contribution). "Active Preference-Based Gaussian Process Regression for Reward Learning", Robotics: Science and Systems (RSS), Corvallis, Oregon, USA, Jul. 2020; doi: 10.15607/rss.2020.xvi.041.
59. Z Cao*, **E Bıyık***, WZ Wang, A Raventos, A Gaidon, G Rosman, D Sadigh (*equal contribution). "Reinforcement Learning based Control of Imitative Policies for Near-Accident Driving", Robotics: Science and Systems (RSS), Corvallis, Oregon, USA, Jul. 2020; doi: 10.15607/rss.2020.xvi.039
60. M Kwon, **E Bıyık**, A Talati, K Bhasin, DP Losey, D Sadigh. "When Humans Aren't Optimal: Robots that Collaborate with Risk-Aware Humans", ACM/IEEE International Conference on Human-Robot Interaction (HRI), Cambridge, United Kingdom, Mar. 2020; doi: 10.1145/3319502.3374832. ***Honorable mention award.***
61. **E Bıyık**, DA Lazar, D Sadigh, R Pedarsani. "The Green Choice: Learning and Influencing Human Decisions on Shared Roads", 58th IEEE Conference on Decision and Control (CDC), Nice, France, Dec. 2019; doi: 10.1109/CDC40024.2019.9030169.
62. C Basu, **E Bıyık**, Z He, M Singhal, D Sadigh. "Active Learning of Reward Dynamics with Hierarchical Queries", IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Macau, China, Nov. 2019; doi: 10.1109/IROS40897.2019.8968522.
63. **E Bıyık**, M Palan, NC Landolfi, DP Losey, D Sadigh. "Asking Easy Questions: A User-Friendly Approach to Active Reward Learning", 3rd Conference on Robot Learning (CoRL), Osaka, Japan, Oct. 2019.
64. **E Bıyık***, J Margoliash*, SR Alimo, D Sadigh (*equal contribution). "Efficient and Safe Exploration in Deterministic Markov Decision Processes with Unknown Transition Models", American Control Conference (ACC), Philadelphia,

Pennsylvania, USA, Jul. 2019; doi: 10.23919/ACC.2019.8815276.

65. **E Bıyık***, DA Lazar*, R Pedarsani, D Sadigh (*equal contribution). "Altruistic Autonomy: Beating Congestion in Shared Roads", 13th International Workshop on Algorithmic Foundations of Robotics (WAFR), Mérida, México, Dec. 2018; doi:
10.1007/978-3-030-44051-0_51
66. **E Bıyık**, D Sadigh. "Batch Active Preference-Based Learning of Reward Functions", 2nd Conference on Robot Learning (CoRL), Zürich, Switzerland, Oct. 2018.
67. HC Baykara*, **E Bıyık***, G Gül*, D Onural*, AS Öztürk*, İ Yıldız* (*equal contribution). "Real-Time Detection, Tracking and Classification of Multiple Moving Objects in UAV Videos", 29th IEEE International Conference on Tools with Artificial Intelligence (ICTAI), Boston, Massachusetts, USA, Nov. 2017; doi: 10.1109/ICTAI.2017.00145.
68. **E Bıyık**, J Barbier, M Dia. "Generalized Approximate Message-Passing Decoder for Universal Sparse Superposition Codes", IEEE International Symposium on Information Theory (ISIT), Aachen, Germany, Jun. 2017; doi: 10.1109/ISIT.2017.8006798.

Workshop Papers

69. A Liang*, Y Korkmaz*, J Zhang*, J Zhang, A Anwar, S Kaushik, Y Wang, Y Xiang, D Held, D Fox, A Gupta, S Tu†, **E Bıyık**† (*equal contribution, †equal advising). "SPUR: Scaling Reward Learning from Human Demonstrations"
    - NeurIPS 2025 Workshop on Embodied World Models for Decision Making (EWM), San Diego, CA, USA, Dec. 2025.
    - 3rd Workshop on Learning Effective Abstractions for Planning (LEAP) at CoRL 2025, Seoul, South Korea, Sep. 2025.
    - CoRL 2025 Workshop on Evaluation and Deployment Across the Robot Learning Lifecycle (Eval&Deploy), Seoul, South Korea, Sep. 2025.
70. Y Tang*, X Xiong*, J Xi, J Li, **E Bıyık**, S Koenig (*equal contribution). "RAILGUN: A Unified Convolutional Policy for Multi-Agent Path Finding Across Different Environments and Tasks"
    - The 18th International Symposium on Combinatorial Search (SoCS 2025), Glasgow, Scotland, United Kingdom, Aug. 2025 (Extended Abstract), doi: 10.1609/socs.v18i1.36015
    - Workshop on Foundation Models and Neuro-Symbolic AI for Robotics (FMNS) at the 2025 International Conference on Robotics and Automation (ICRA), Atlanta, Georgia, USA, May 2025
71. J Zhang*, Y Luo*, A Anwar*, SA Sontakke, JJ Lim, J Thomason, **E Bıyık**, J Zhang (*equal contribution). "ReWiND: Language-Guided Rewards Teach Robot Policies without New Demonstrations"
    - 2nd Workshop on Out-of-Distribution Generalization in Robotics at RSS 2025, Los Angeles, California, USA, Jun. 2025 ***(Best paper award)***
    - RSS 2025 Workshop on Learned Robot Representations (RoboReps), Los Angeles, California, USA, Jun. 2025 ***(Best paper award nominee)***
    - Inductive Biases in Reinforcement Learning Workshop at RLC 2025, Edmonton, Alberta, Canada, Aug. 2025
    - RSS 2025 Large Foundation Model for Interactive Robot Learning Workshop, Los Angeles, California, USA, Jun. 2025
72. M Hwang, Y Korkmaz, D Seita†, **E Bıyık**† (†equal advising). "Causally Robust Preference Learning with Reasons"
    - RSS 2025 Workshop on Human-in-the-Loop Robot Learning: Teaching, Correcting, and Adapting, Los Angeles, California, USA, Jun. 2025
    - RSS 2025 Workshop on Continual Robot Learning from Humans, Los Angeles, California, USA, Jun. 2025
73. Y Ling*, K Owalekar*, O Adesanya, **E Bıyık**, D Seita (*equal contribution). "IMPACT: Intelligent Motion Planning with Acceptable Contact Trajectories via Vision-Language Models"
    - 2nd Workshop on Semantic Reasoning and Goal Understanding in Robotics (SemRob) at RSS 2025, Los Angeles, California, USA, Jun. 2025
    - RSS 2025 Workshop on Large Foundation Models for Interactive Robot Learning, Los Angeles, California, USA, Jun. 2025
    - R3: Reasoning for Robust Robot Manipulation in the Open World Workshop at RSS 2025, Los Angeles, California, USA, Jun. 2025
74. M Hong*, A Liang*, KJ Kim, HB Rajaprakash, J Thomason†, **E Bıyık**†, J Zhang† (*equal contribution, †equal advising). "HAND Me the Data: Fast Robot Adaptation via Hand Path Retrieval", RSS 2025 Workshop on Human-in-the-Loop Robot Learning: Teaching, Correcting, and Adapting, Los Angeles, California, USA, Jun. 2025.
75. P Czempin, R Freedman, E Novoseller, VJ Lawhern, C Allen, **E Bıyık**. "Improving Reward Learning by Estimating Annotator Expertise", RSS 2025 Workshop on Continual Robot Learning from Humans, Los Angeles, California, USA, Jun. 2025.
76. A Liang, G Tennenholtz, C Hsu, Y Chow, **E Bıyık**, C Boutilier. "DynaMITE-RL: A Dynamic Model for Improved Temporal Meta-Reinforcement Learning", Workshop on Automated Reinforcement Learning: Exploring Meta-Learning, AutoML, and LLMs at 41st International Conference on Machine Learning (ICML), Vienna, Austria, Jul. 2024.

77. A Liang, P Czempin, Y Zhou, S Tu, **E Bıyık**. "In-Context Generalization to New Tasks From Unlabeled Observation Data", Workshop on In-Context Learning at 41st International Conference on Machine Learning (ICML), Vienna, Austria, Jul. 2024.
78. J Zhang, Z Liu, M Heo, **E Bıyık**, JJ Lim, Y Liu, R Fakoor. "EXTRACT: Efficient Policy Learning by Extracting Transferrable Robot Skills", Embodied AI Workshop at the IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR), Seattle, Washington, USA, Jun. 2024.
79. J Tien*, Z Yang*, M Jun, SJ Russell, A Dragan, **E Bıyık** (*equal contribution). "Optimizing Robot Behavior via Comparative Language Feedback", 3rd Workshop on Human-Interactive Robot Learning (HIRL) at ACM/IEEE International Conference on Human-Robot Interaction (HRI), Boulder, Colorado, USA, Mar. 2024.
80. A Liang, J Thomason, **E Bıyık**. "ViSaRL: Visual Reinforcement Learning Guided by Human Saliency", Pretraining for Robotics (PT4R) Workshop at the 2023 International Conference on Robotics and Automation (ICRA), London, United Kingdom, May 2023.
81. **E Bıyık**. "Learning from Humans for Adaptive Interaction", 17th Annual Human-Robot Interaction Pioneers Workshop (HRI Pioneers), Sapporo, Hokkaido, Japan, Mar. 2022.
82. M Kwon, **E Bıyık**, A Talati, K Bhasin, DP Losey, D Sadigh."When Humans Aren't Optimal: Robots that Collaborate with Risk-Aware Humans", Cooperative AI NeurIPS Workshop 2021, Virtual, Dec. 2021.
83. **E Bıyık**, A Lalitha, R Saha, A Goldsmith, D Sadigh. "Partner-Aware Algorithms in Decentralized Cooperative Bandit Teams", Artificial Intelligence for Human-Robot Interaction Symposium (AI-HRI) at AAAI Fall Symposium Series, Washington DC, USA, Nov. 2021.
84. **E Bıyık**, A Talati, D Sadigh. "APReL: A Library for Active Preference-based Reward Learning Algorithms", Artificial Intelligence for Human-Robot Interaction Symposium (AI-HRI) at AAAI Fall Symposium Series, Washington DC, USA, Nov. 2021.
85. M Beliaev*, WZ Wang*, DA Lazar, **E Bıyık**, D Sadigh, R Pedarsani (*equal contribution). "Emergent Correlated Equilibrium through Synchronized Exploration", RSS 2020 Workshop on Emergent Behaviors in Human-Robot Systems, Corvallis, Oregon, USA, Jul. 2020.

Ph.D. Dissertation

86. **E Bıyık**. "Learning Preferences for Interactive Autonomy", Ph.D. Dissertation, Department of Electrical Engineering, Stanford University, May 2022.

Preprints & Technical Reports

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